Jevi Waugh

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Experience

The University of Queensland — **School of Electrical Engineering and Computer Science, St Lucia**
Software Engineering Demonstrator — *July 2025*
Delivered content for CSSE1001, guide students in programming fundamentals, debugging, object-oriented design, and software engineering practices. Provide feedback to improve coding skills and confidence.

R.M. Williams — ***Sales associate, Chermside — April - October 2024***
Delivered customer service, performed boot repairs, managed logistics and deliveries, applied AI-assisted merchandising, and assumed leadership responsibilities.

Queensland AI Safety Initiative (QuASI) — ***Technical Fellowship, St. Lucia — Summer 2025***
Completed an 8-week technical fellowship on AI safety, exploring Mechanistic Interpretability, Reinforcement Learning, Goal Misgeneralisation, and broader Alignment challenges. Previously earned an Artificial Intelligence Alignment Reading Course certificate in 2021.

Defence Families community — ***Volunteer, Secret Harbour — 2021***
Volunteered in the Defence School Mentor Program, assisting with ANZAC and Remembrance Day events through displays, poster design, and speeches that honored servicemen and women and fostered community remembrance. Defence Work Experience Certificate & Defence Technologies Extension Award were earned during that time

Projects

Hugging Face, Google Collab, OpenCV
Internal Dynamics of Vision Transformer Architectures — *July 2025 – current*
This project examines internal operational dynamics within Vision Transformer architectures. The work centers on capturing and processing attention distributions across sequential processing blocks and functional units. A methodology was established to extract these internal signals from a pretrained DeiT model in a non-invasive manner. Further work will investigate direct modifications to the attention formulation to evaluate how structural changes affect representational behavior. In collaboration with Dr. Kasra Khosoussi.

DECO3801 Capstone project - TraceForest and StyleCodeGPTSensor — ***Jira, PyTorch, scikit-learn***
Developed an explainable Random Forest model achieving 94% accuracy to detect AI-assisted code using stylometric AST features. Integrated SHAP-based explainability to identify which stylometric attributes influenced AI vs human predictions. Implemented model adapters for both Python and Java, returning label probabilities and contributing features for bot PR review comment within secure development workflows.Upgraded CodeGPTSensor by integrating stylometric features through transfer learning/fine-tuning of the pretrained encoder for enhanced AI-assisted code detection above 95% accuracy.

DAWNBench Challenge — ***PyTorch, CUDA, Google Colab, CIFAR-10***
Developed and trained a ResNet-style convolutional neural network in PyTorch achieving over 93% test accuracy on CIFAR-10 within 30 minutes. Implemented data augmentation, mixed-precision training, cosine learning-rate scheduling, and efficient GPU utilization for speed optimization.

OASIS Challenge – UNet Brain MRI Segmentation — ***PyTorch, CUDA, Google Colab***
Implemented a 2D UNet for multi-class brain MR segmentation with one-hot outputs; trained with Cross-Entropy and Dice loss with early stopping; achieved over 90% Dice Similarity Coefficient (DSC) on all labels; ran inference and visualized results on Google Colab using an NVIDIA A100 GPU (Data was licensed by UQ).

Persistent Homology ML Pipeline (PH-AML) — ***Julia, Python, Jupyter Notebook***
Developing a pipeline integrating topological data analysis into supervised learning models by extracting persistent homology features (barcodes, landscapes) from high-dimensional data using Vietoris–Rips filtrations, enhancing classification between K-complexes.

Education and Qualifications

The University of Queensland — **St Lucia, Brisbane**
Bachelor of Computer Science in Machine Learning — *2024 - Current*

Curtin College — **Bentley, Perth**
Diploma in Information Technology | Graduated in 2022 with 6.0 GPA — *Feb 2022 – Oct 2022*

References

Senior Lecturer, UQ
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